

Four-Level SapBERT: Improvement Study

Revised comparison | Levels 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent

This improvement study uses the same 10,000 supplied records, frozen SapBERT embeddings and original grouped partitions. It expands the search to PCA-64, PCA-128 and PCA-256, stronger/weaker regularization, class balancing, PCA whitening and structured-only controls. No labels are changed.

Selected: SapBERT + PCA-128 + Logistic Regression

Training settings: C=100; balanced class weights.

Patient preprocessing includes quadratic numeric terms; the classifier family and frozen SapBERT encoder are unchanged.

Selected total input size: 178 features. PCA retained variance: 95.51%.

Retrospective metric (%)	Initial	Prior round	Selected
Accuracy	85.19	85.99	86.29
Macro precision	85.71	86.52	86.75
Macro recall	85.90	86.64	86.87
Macro F1	85.80	86.58	86.80
Emergency recall	92.16	92.16	91.42
Under-triage	7.60	6.70	6.80

Selected quadratic weighted kappa: 0.9375; mean absolute level error: 0.1376; over-triage: 6.90% (initial 7.20%).

Mean development CV macro F1 change versus initial: +0.59 percentage points. Selection uses all five development folds and an emergency-recall constraint; the test results above do not select the winner.

Paired development bootstrap versus initial: F1 gain 95% interval +0.04 to +1.13 percentage points. This conditional interval excludes zero. This diagnostic excludes model-selection uncertainty.

Evaluation scope: These 1,999 test rows were already examined in the previous study. The new scores are retrospective comparisons, not a fresh independent estimate. New labelled records are needed to confirm generalization. Saved concepts are evaluated here; live translation accuracy is not measured.

Trade-offs remain visible: compare emergency recall and under-triage as well as aggregate scores. SapBERT is not fine-tuned; these gains do not establish clinical validity.

Complete Development Cross-Validation

All 49 configurations; 245 fits across five grouped folds. F1, accuracy and emergency recall are percentages. SD is the fold-to-fold F1 standard deviation in percentage points. The incumbent is lr_pca64_c10_balanced1.

Configuration	F1	SD	Accuracy	Emergency recall
lr_quad128_c100	86.57	0.70	86.09	92.65
lr_pca128_c100_balanced0	86.57	0.59	86.05	91.60
lr_refine128_c300	86.50	0.75	86.00	92.52
lr_pca128_c1000_balanced1	86.48	0.69	85.98	92.28
lr_quad128_c300	86.48	0.68	85.99	92.34
lr_pca128_c100_balanced1	86.46	0.58	85.98	92.65
lr_quadratic64_c100_balanced1	86.41	0.80	85.93	92.83
lr_pca128_c10_balanced0	86.39	0.57	85.84	91.73
lr_refine128_c30	86.32	0.51	85.81	92.89
lr_quadratic64_c100_balanced0	86.31	0.51	85.81	91.05
lr_refine64_c30	86.26	0.61	85.75	93.50
lr_pca128_c1000_balanced0	86.24	0.47	85.71	91.05
lr_pca256_c10_balanced0	86.24	0.44	85.68	91.67
lr_quadratic64_c10_balanced0	86.23	0.66	85.71	91.24
lr_quadratic64_c10_balanced1	86.21	0.52	85.71	93.08
lr_pca64_c1000_balanced0	86.19	0.54	85.68	90.93
lr_pca64_c100_balanced1	86.19	0.46	85.69	93.01
lr_quad256_c100	86.18	0.92	85.66	91.54
lr_pca64_c100_balanced0	86.17	0.78	85.65	91.05
lr_pca128_c10_balanced1	86.16	0.43	85.63	93.07
lr_refine64_c300	86.06	0.52	85.55	92.83
lr_pca256_c100_balanced0	86.06	0.57	85.50	90.75
lr_pca256_c10_balanced1	86.05	0.46	85.51	92.95
lr_pca64_c1000_balanced1	86.04	0.55	85.53	92.77
lr_pca64_c10_balanced1	85.98	0.53	85.46	93.38
lr_pca256_c100_balanced1	85.91	0.70	85.36	91.97
lr_quad256_c300	85.90	1.07	85.35	91.24
lr_pca64_c10_balanced0	85.86	0.61	85.31	91.48
lr_pca256_c1000_balanced1	85.14	0.76	84.54	90.99
lr_pca256_c1000_balanced0	85.12	0.67	84.50	90.32
lr_pca128_whiten	85.06	0.46	84.51	92.34
lr_pca64_whiten	85.04	0.57	84.54	92.09
hgb_refine64_leaf7	84.76	0.80	84.26	90.38
hgb_fused_64_leaf7	84.34	0.65	83.81	91.36
hgb_fused_64_leaf15	84.24	0.76	83.74	90.81
hgb_fused_128_leaf15	83.75	0.59	83.23	90.44
lr_pca256_whiten	83.49	0.82	82.89	90.56
hgb_refine64_leaf31	83.45	0.63	82.94	90.14
hgb_fused_128_leaf7	83.41	0.98	82.83	91.61
lr_quadratic64_c1_balanced1	83.21	0.55	82.64	93.02
lr_quadratic64_c1_balanced0	83.20	0.73	82.56	90.26
rf_refine64_leaf1	79.21	0.34	79.13	95.96
rf_refine64_leaf3	78.31	0.42	78.33	96.63
rf_fused_64	78.31	1.21	77.82	87.68
rf_refine128_leaf3	76.48	0.76	76.72	96.63
rf_refine128_leaf1	76.48	0.85	76.64	95.77
hgb_structured_0_leaf7	76.23	1.18	75.55	87.38
hgb_structured_0_leaf15	75.87	1.00	75.15	86.40
rf_structured_0	75.50	0.73	74.85	86.03

Selection: highest mean macro F1 among candidates whose mean emergency recall is within one percentage point of the incumbent; accuracy resolves ties. A higher F1 does not qualify a candidate whose emergency recall falls below the threshold. This is not a clinical safety guarantee.

Original 768-D versus PCA-64 Baselines

All original fixed comparisons are retained below. They use the same 1,999 previously examined test rows. Values are percentages; precision, recall and F1 are macro-averaged. These baseline runs are from the first round and are not new experiments.

Text only

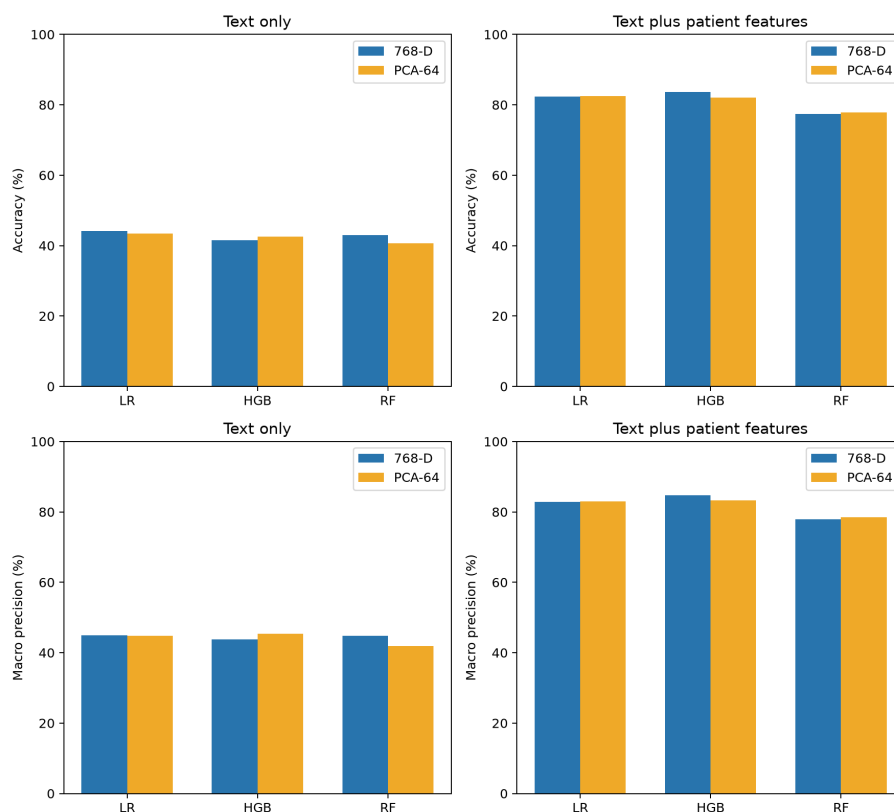
Text size	Classifier	Accuracy	Precision	Recall	F1
768	Logistic Regression	44.17	44.89	44.87	43.95
768	HistGradientBoosting	41.52	43.72	40.95	41.70
768	Random Forest	42.92	44.82	42.93	43.49
64	Logistic Regression	43.42	44.74	44.20	43.26
64	HistGradientBoosting	42.57	45.34	41.85	42.80
64	Random Forest	40.62	41.88	40.93	41.08

Text plus patient features

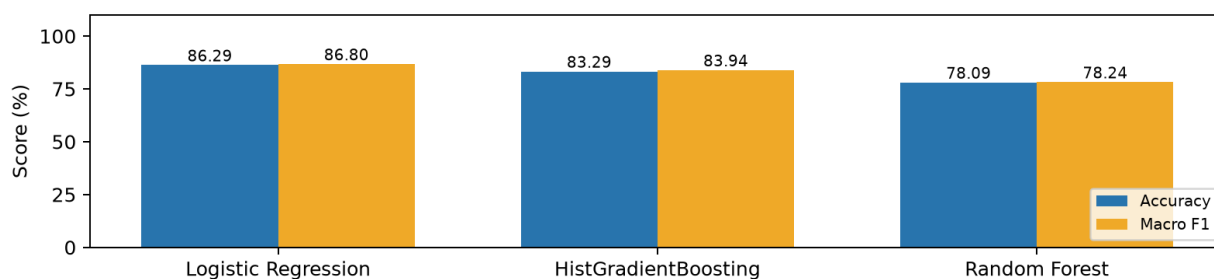
Text size	Classifier	Accuracy	Precision	Recall	F1
768	Logistic Regression	82.29	82.89	83.10	82.92
768	HistGradientBoosting	83.64	84.72	84.03	84.21
768	Random Forest	77.44	77.87	78.24	77.93
64	Logistic Regression	82.44	83.01	83.22	83.03
64	HistGradientBoosting	82.09	83.28	82.54	82.76
64	Random Forest	77.89	78.47	78.65	78.35

The original fused PCA-64 input has 86 features: 64 text components and 22 encoded patient features. The full fused input has 790 features. All training transformations are fitted without validation/test rows.

Performance Graphs



Tuned classifiers: retrospective comparison

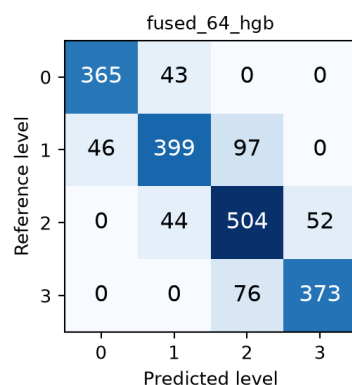
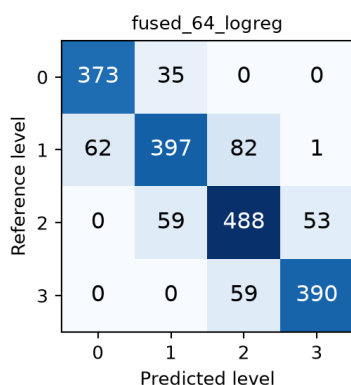
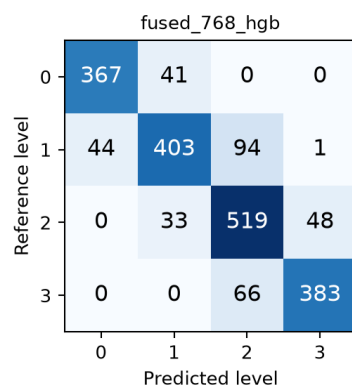
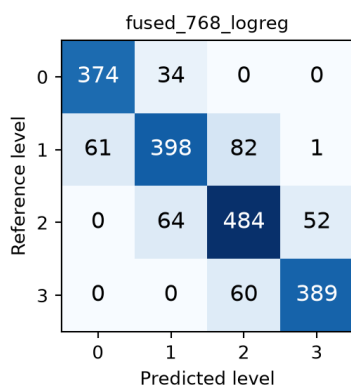
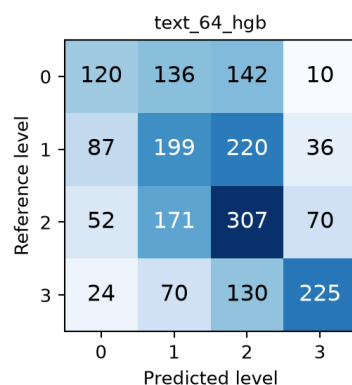
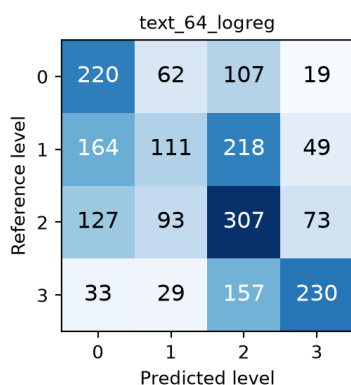
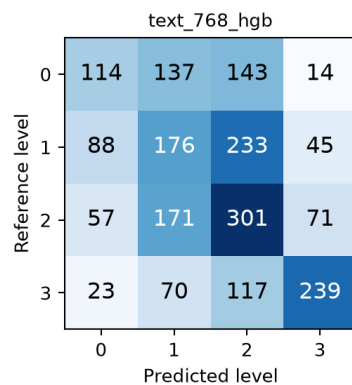
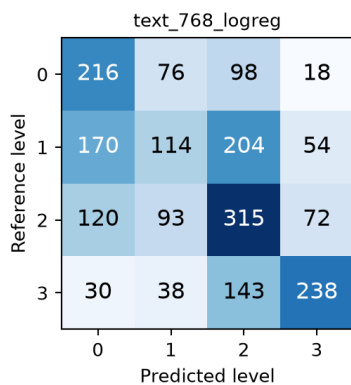


Classifier	Accuracy	Precision	Recall	F1
Logistic Regression	86.29	86.75	86.87	86.80
HistGradientBoosting	83.29	84.13	83.82	83.94
Random Forest	78.09	79.88	78.83	78.24

Each family uses its highest fused development CV F1 setting. These descriptive choices do not override the emergency-recall constraint for deployment. Scores are percentages; precision, recall and F1 are macro averages.

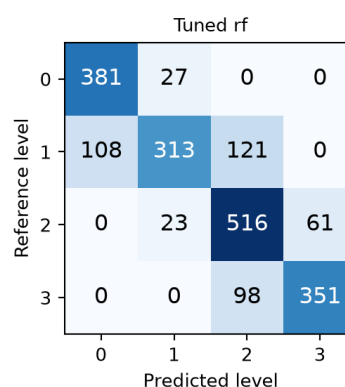
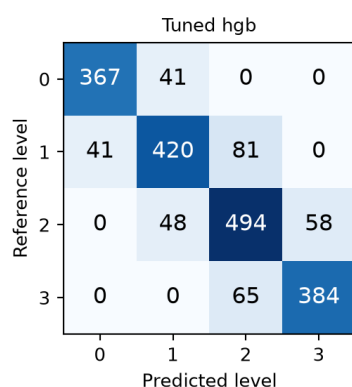
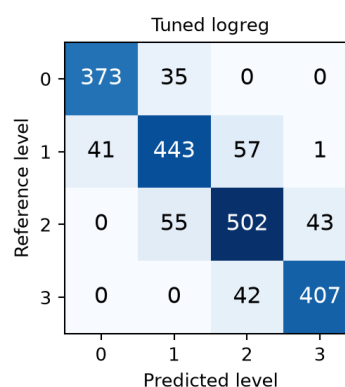
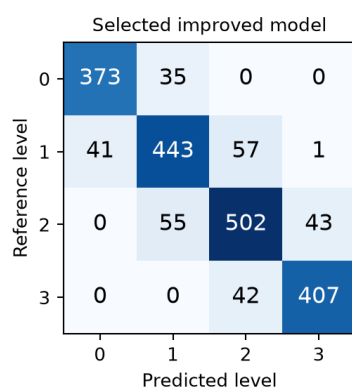
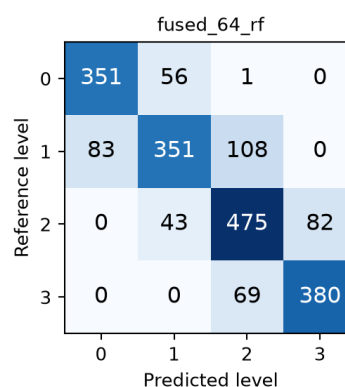
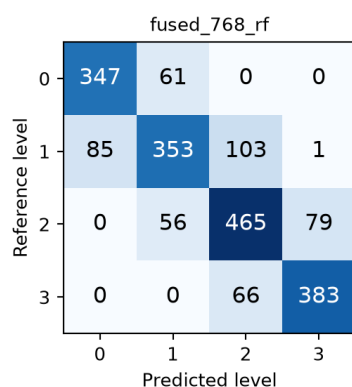
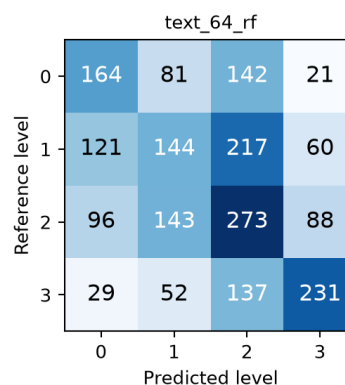
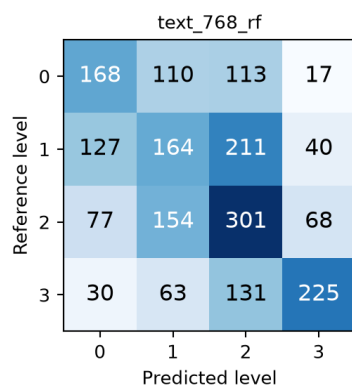
Blue and yellow bars preserve the original comparison style. The selected improved result is shown separately on page 1 because it follows a larger development search.

Baseline Confusion Matrices: LR and HGB



Rows are reference labels; columns are predictions. Level order: 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent. Baseline matrices are unchanged from round one.

Baseline and Selected Confusion Matrices



Rows are reference labels; columns are predictions. Level order: 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent. Baseline matrices are unchanged from round one.

Class Results and Error Review

Level	Precision (%)	Recall (%)	F1 (%)	Support
0 Emergency	90.10	91.42	90.75	408
1 Urgent	83.11	81.73	82.42	542
2 Standard	83.53	83.67	83.60	600
3 Non-urgent	90.24	90.65	90.44	449

Development out-of-fold confusion matrix

Reference / predicted	0	1	2	3
0	1512	120	0	0
1	183	1752	234	3
2	0	212	1985	203
3	0	0	158	1639

An input audit found no identical development input combinations with conflicting targets. This does not establish that labels are clinically correct. A local review CSV lists development errors and model confidence; it is not committed because it contains individual records.

Use the review list to investigate unclear label boundaries, missing symptom severity/duration/history and data entry issues. Do not replace reference labels with model predictions merely to improve agreement.

Literature context (different task)

Model	Accuracy (%)	Recall (%)	F1 (%)
TF-IDF + LR	75.00	98.80	54.40
TF-IDF + RF	75.10	96.40	58.20
BiLSTM	74.60	84.60	67.00
CNN	72.30	78.70	66.70
Current four-level SapBERT	86.29	86.87	86.80

Seo et al., Scientific Reports (2025), Table 2. DOI: 10.1038/s41598-025-99874-0. Different binary KTAS task; publication metric definitions differ. Context only, not evidence of superiority over these models.

Methods, Reproducibility and Limits

Data and preprocessing

The original 8,001 development / 1,999 test partition and five grouped folds are reused. Groups do not overlap across training and validation/test boundaries. Input and embedding hashes are checked before and after the run. Median imputation, categorical encoding, scaling and PCA are learned inside each development fold. No label-derived columns are model inputs.

Encoder and search

SapBERT-from-PubMedBERT-fulltext uses frozen 768-D CLS embeddings, L2 normalization, a 64-token limit and the original pinned revision. Logistic Regression varies $C=10, 100$ and 1000 with and without balancing, plus whitened PCA with $C=0.1$. Six additional LR configurations use quadratic numeric terms, with $C=1/10/100$ and optional balancing. HGB tests 7/15 leaves with regularization; Random Forest tests structured and fused inputs. Every candidate is evaluated on all five folds.

The refinement adds 14 configurations: balanced LR $C=30/300$ at PCA-64/128; quadratic LR $C=100/300$ at PCA-128/256; HGB with 7/31 leaves and 800/400 iterations; and RF with 500 trees, square-root feature sampling and leaf sizes 1/3. Classifier families remain unchanged.

Selection and deployment

The incumbent is included in the same search. Selection is recorded before the retrospective test evaluation. The exported model must reproduce saved predictions through the shared GUI/CLI preprocessing. Model artefacts and the incumbent are retained separately. Quadratic terms, if selected, are applied by the fitted numeric preprocessing pipeline. No label correction or encoder fine-tuning is claimed.

Research interpretation

The larger search can overfit development cross-validation. The old test set is already exposed, so its scores must not be presented as untouched confirmation. A new independently labelled dataset is needed for confirmation. Supplied/recovered concepts bypass live translation during this evaluation. The provider could not supply label-assignment rules; independent per-record review is not documented. These are research comparisons, not clinical validation.

Files

protocol.json freezes candidates and source identities; cross_validation.csv contains every fit; cv_summary.csv reports every candidate; selection.json records the decision; retrospective_results.json contains class scores and matrices. Source data, embeddings and per-record error files remain local.